cost effectiveness to the EV sector, but it still needs support in terms of governance, which can help wider deployment and faster adoption.

China is now adopting the EV culture in a big way, supported by government initiatives like restrictions on the use of conventional fuel engine vehicles in certain major cities, a step taken in the public interest to reduce pollution. Allied industries like the manufacturers of batteries, components and electronic accessory manufacturers have been given tax holidays to catalyse this growth. These are bold moves that are expected to drive exponential growth in the segment, setting an example for others to follow. One major move made by China is in public transport. Today, more than 200,000 buses on China's roads are electrically powered, and 20 per cent of all new orders annually are being equipped with EV features.

After the climate change conference held at Morocco in 2016, it is widely believed that EVs are the future for a sustainable transport system, and that's why there is a global impetus to shift public transport systems towards optimal adoption of EVs. There is a consensus that by 2020, twenty million cars globally must be electrically powered—this is being championed under the Electric Vehicles Initiative (EVI). Prior to the Morocco convention, the Paris Declaration on electro–mobility and climate change focused on the fact that transport contributes almost 23 per cent to greenhouse gas (GHG) emissions, and this is increasing at a faster rate than in most sectors. If the right steps are not taken in time, it could contribute to 50 per cent of GHG emissions by 2050. This will lead to rising temperatures and global imbalances. Are we ready to counter this trend?

Therefore it is advisable for governments to invest in sustainable energy sources that can impact human life in a positive way. The steps that can lead to this transition are: a dramatic change in the transportation portfolio, with a set target to replace fossil fuel powered vehicles with EVs; investment in global research to understand the dynamics across temperature zones; incentives to manufacturing industries with EVs in their product portfolio; and creation and implementation of global agreements as well as policy frameworks to drive change.

There are various encouraging government initiatives, too, for this sector. The Indian government has set up a Green Urban Transport Scheme (GUTS) with an initial investment of Rs 250 billion. This will help propel green energy initiatives and infrastructure at the national, state and city levels, all of which will impact the EV market. Another promising policy initiative is the National Electric Mobility Mission – 2020, mandated under the FAME scheme. This is

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India today is at a stage in the EV adoption lifecycle that is similar to where we were with computers and the Internet during the 1980s and 90s. We could call it the take-off point. There is a huge opportunity for us to leapfrog ahead of the early adopters due to the sheer power of our buyer base. What we need today is the willingness to try and explore the ways in which we can increase the adoption of EVs in the country. We need to check what our strengths are in terms of technology and provide an ideal investment climate. We also need to harvest home-grown talent as well as give the foreign brands a level playing field, to encourage them to venture into this space under the Make in India platform.

The introduction of the 'Faster Adoption and Manufacturing of Hybrid and Electric Vehicles' or FAME scheme by the government in 2015, to promote hybrid vehicles and EVs, is a step in the right direction. And with governments in major states thinking of replacing their fleet vehicles with EVs, there are some things to cheer about.

India is currently estimated to have 500,000 electric two-wheelers, about 200,000 e-rickshaws, 100,000 e-powered buses and a few thousand e-cars, which is a miniscule slice of the entire Indian auto industry. As per the Society of Indian Automobile Manufacturers (SIAM) data, the auto sector today contributes to 7.1 per cent of India's GDP, with the two-wheeler segment accounting for 81 per cent of the growth. So, what are the factors that are expected to change the dynamics of the automobiles industry and the EV sector in India, in particular? Here is a list.

- Global investments in India: Tesla, a global giant in the EV segment, and South Korea's Kia Motors currently want to enter India with their EVs. KIA has gone a step further and narrowed down its choices for a potential manufacturing location to Maharashtra or Andhra Pradesh.
- Toyota and Nissan also have plans of coming in with a larger bouquet of offerings.
- Suzuki Motor Corp is eyeing the lithium-ion battery segment, and is planning a JV with Denso Corp and Toshiba Corp to produce batteries for the Indian market.
- Hero Motor Corp is eyeing the EV segment and has invested US$ 31 million to acquire a controlling stake in Bengaluru based tech firm, Ather Energy, which is the brain child of two IIT Madras students (who were covered under the government's FAME-2015 scheme).
- Ford Motors is planning to invest US$ 200 million in India, to build a global technology and business centre in Chennai. This will be a product design hub that will propel the EV segment and other mobility services across global markets.
- Ola Cabs, one of India's leading car rental service companies, is planning to introduce one million EVs in collaboration with an Indian manufacturer within the next few years starting 2018 – this has been reported in the investment report of Softbank Group, investment partners to Ola Cabs.